Native N83 retained-base diagnostics

19 October 2026 · run-20261019T0140Z

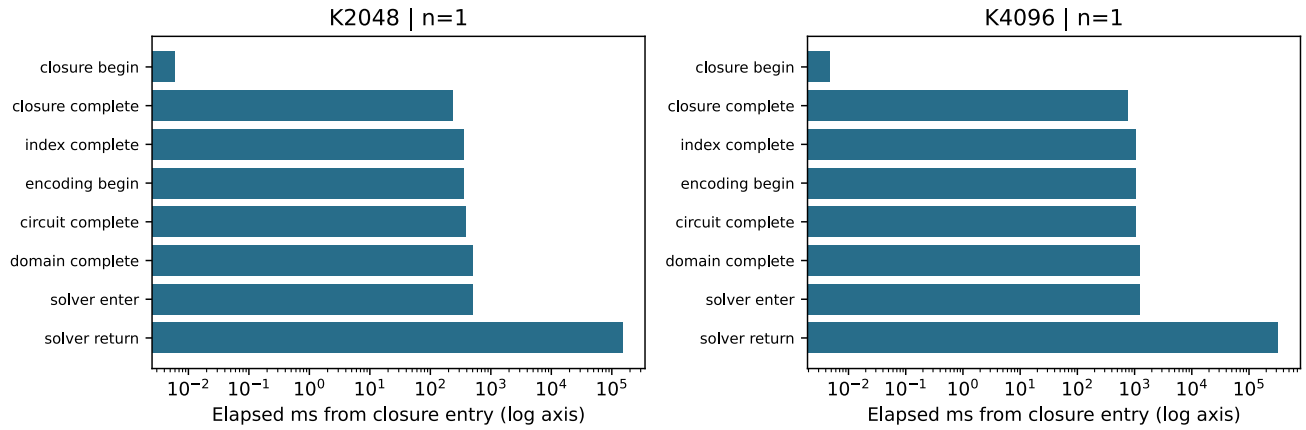
The approved historical ZIP is uploaded and fresh-GET verified: 5,771,044 bytes, 224 entries. Its immutable SHA256 is `b7afa24e3ca80630344b5e584915227c5e177c66068f6f5749ce72ba4894ba92`. The publication addendum supersedes the old pending-upload state; frozen history is unchanged.

Observed: Both corrected natural arms returned at 1,000 conflicts with zero models, zero relations and rank zero. Independent public-query replay and cleanup passed. Outer wall: K2048 163378.646436 ms; K4096 337901.101152 ms. This resolves bounded terminal observability; it establishes no cold-runtime winner.

Method: Public a=0 Koblitz GF(2^83) fixtures; exact K2048/K4096 public_x_hash retained bases, seed 2026100801. Each corrected v3 arm: one unpinned natural m5 orbit-chain-S3 trial, 1000 conflicts, 600 s wall, 32 models, 150000 variables, 2000000 domain clauses. Independent native arithmetic replays the actual aG+bQ query. Native cards pin binaries, source, inputs and commands before execution.

Resources and accounting: Docker 1 CPU/4 GiB/no swap/network none/64 PIDs/read-only inputs and root/task UID/no capabilities/no-new-privileges. Independent inner hard timeout, live/final inspection and cleanup. L0 diagnostic clocks; no A/A interval or timing isolation. Build/test activity on host is separate overhead. Outer process wall and closure-relative phase clocks have different origins.

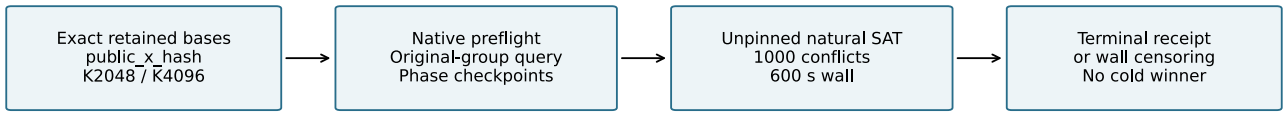
Recorded phase checkpoints | native corrected v3 arms



solver enter includes preprocessing; absence of a return leaves final counters unknown

Recorded checkpoints, n=1 per arm; no uncertainty interval. solver_enter includes preprocessing. A missing solver_return cannot supply final counters. Data is copied from native audits; elapsed checkpoints are not exclusive phase durations.

Corrected v3 execution; earlier v1/v2 outcomes preserved separately



Fully specified selector witness checks encoding consistency; free-selector N83 control remains censored

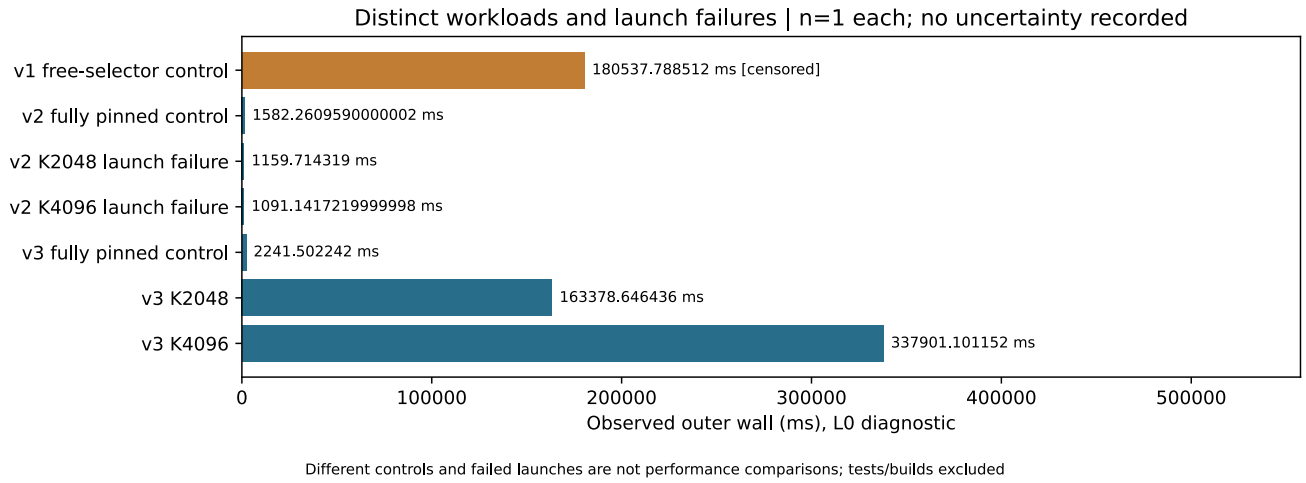
No natural-yield estimate, fastest base, fully charged cold IC runtime, external-tool comparison, matched rho comparison, asymptotic claim or nonexistence conclusion follows.

Controls, failures and verification

V1 small-field m5/m6 ordered/unordered assertions passed with 4 SAT returns. Its original N83 fixed-coordinate/free-selector control reached fixed_units_installed at 113.114676 ms, then was censored at 180537.788512 ms outer wall. It failed its positive gate; original natural cards remained unlaunched. Final counters are unknown.

V2 explicitly introduced a different control: known coordinates, representatives and phases fixed on the same full 1182/866 domains. SAT model, independent group sum and selector/phase replay passed with 0 conflicts. This is encoding consistency only; the free-selector N83 witness-search gap remains. V3 repeats this control on its newly pinned runner. A coherent custody update did not hide a forged phase: independent replay rejected it.

Both v2 retained-arm launches failed before search: the runner requested public_x_sequential from public_x_hash manifests. Both errors and cleanup records remain. V3 corrects the policy and adds a native manifest preflight, separately frozen with new specs/cards. No failed launch is counted as a search timeout or omitted from accounting.



Distinct workloads are shown for custody/accounting, never as a speed comparison. The v1, v2 and v3 envelopes are additional, not reset budgets. See the Markdown report for exact final statuses, native audits and source-linked receipts.

Validation: Fresh measured-source release run exited 0: 4397 library tests passed, 118 ignored; 19 exporter tests passed; fixed-witness example 0 tests. New selector locator test: 1 passed; native runner/card tests: 13 passed. All 92 historical compact-catalog files re-hashed unchanged. Forged K2048 terminal conflicts and coherently rehashed public-query coefficients were independently rejected. Post-merge integration validation is recorded separately.

Revisions: V1 fc7d1a63f75add58b9eb619cad93c730d5b047c0; v2 c259497ec9cf1fde358ffc91eabe8c2cff242f95; v3 27d17bd17b9fe9c6b4ce04e0c9206706b2b5c804. Full commands, raw logs, receipts, Docker inspections, source snapshots, frozen binaries and inputs accompany the archive; compact evidence is under verification/native-phase-20261019-v3. REPORT.md provides links, data and editable rendering source.

Remaining: Full finite-grid sweep, free-selector N83 known-witness search, useful natural rank, column logs, individual logs, native positive-relation/matrix replay and full cold comparison remain open. Censoring is not UNSAT. Canonical leaderboard/scoreboard/progress-timeline checked unchanged: no qualified gain or winner is established.